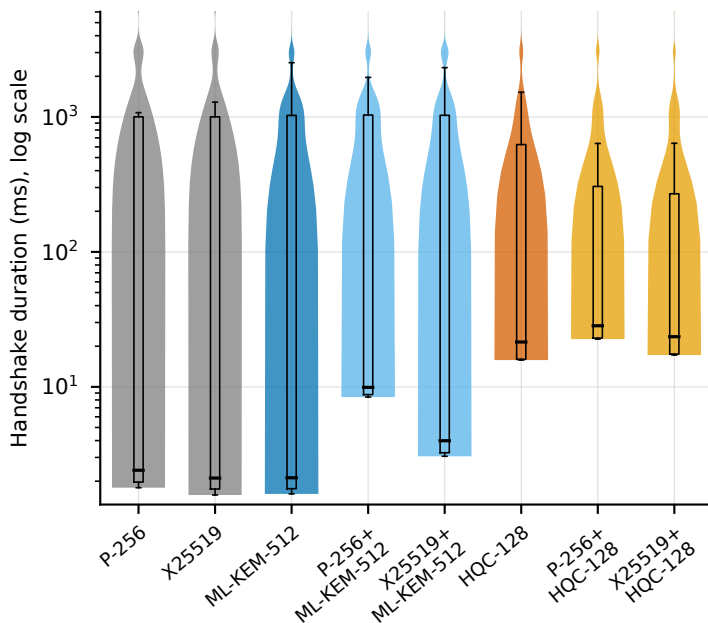
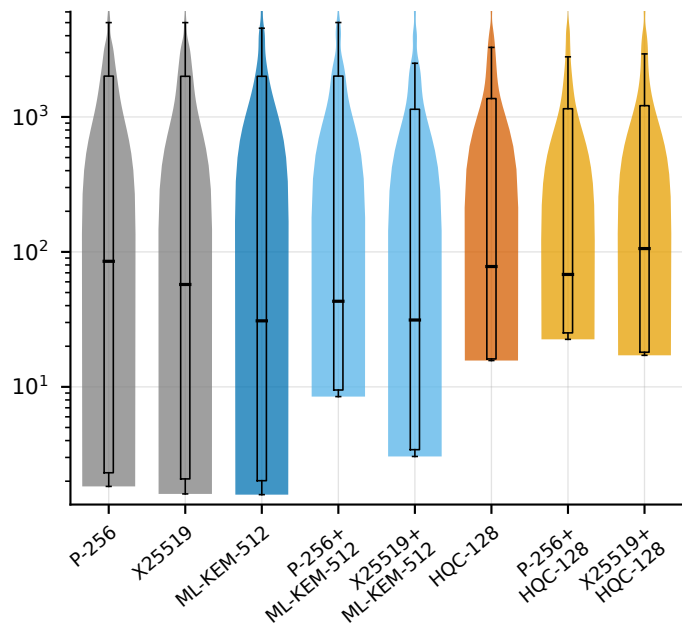


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	1.05k	181.9	10.00k	28/500	1.26k	163.1	13.22k	16/500
X25519	1.01k	192.7	11.00k	21/500	1.20k	163.8	14.37k	11/500
ML-KEM-512	717.0	193.6	10.06k	9/500	1.33k	183.5	16.43k	12/500
P-256+ML-KEM-512	700.0	193.1	11.15k	12/500	1.22k	178.9	16.89k	6/500
X25519+ML-KEM-512	758.8	189.1	9.58k	14/500	1.05k	184.5	13.88k	9/500
HQC-128	599.7	230.5	12.40k	9/500	1.17k	190.1	19.49k	9/500
P-256+HQC-128	500.7	233.1	9.76k	2/500	1.00k	177.8	9.92k	10/500
X25519+HQC-128	450.4	233.4	10.48k	4/500	1.10k	175.8	11.86k	4/500